

13-4a The Relationship between Short-Run and Long-Run Average Total Cost

For many firms, the division of total costs between fixed and variable costs depends on the time horizon. Consider, for instance, a car manufacturer such as Ford Motor Company. Over a period of only a few months, Ford cannot adjust the number or sizes of its car factories. The only way it can produce additional cars is to hire more workers at the factories it already has. The cost of these factories is, therefore, a fixed cost in the short run. By contrast, over a period of several years, Ford can expand the size of its factories, build new factories, or close old ones. Thus, the cost of its factories is a variable cost in the long run.

Because many decisions are fixed in the short run but variable in the long run, a firm's long-run cost curves differ from its short-run cost curves. [Figure 6](#) shows an example. The figure presents three short-run average-total-cost curves—for a small, medium, and large factory. It also presents the long-run average-total-cost curve. As the firm moves along the long-run curve, it is adjusting the size of the factory to the quantity of production.

Figure 6

Average Total Cost in the Short and Long Runs

Because fixed costs are variable in the long run, the average-total-cost curve in the short run differs from the average-total-cost curve in the long run.

This graph shows how short-run and long-run costs are related. The long-run average-total-cost curve has a much flatter U-shape than the short-run average-total-cost curve. In addition, all the short-run curves lie on or above the long-run curve. These properties arise

because firms have greater flexibility in the long run. In essence, in the long run, the firm gets to choose which short-run curve it wants. But in the short run, it has to use whatever short-run curve it has, as determined by decisions it has made in the past.

The figure shows an example of how a change in production alters costs over different time horizons. When Ford wants to increase production from 1,000 to 1,200 cars per day, it has no choice in the short run but to hire more workers at its existing medium-sized factory. Because of diminishing marginal product, average total cost rises from \$10,000 to \$12,000 per car. In the long run, however, Ford can expand both the size of the factory and its workforce, and average total cost returns to \$10,000.

How long does it take a firm to get to the long run? The answer depends on the firm. It can take a year or more for a major manufacturing firm, such as a car company, to build a larger factory. By contrast, a person running a coffee shop can buy another coffee maker within a few days. There is, therefore, no single answer to the question of how long it takes a firm to adjust its production facilities.

Chapter 13: The Costs of Production: 13-4a The Relationship between Short-Run and Long-Run Average Total Cost

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Printed By: Mike Benton (mbento6@wgu.edu)

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